

Lecture 6: Atmospheres II

Clouds, Weather, & Climate

Planetary Systems

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Learning objectives

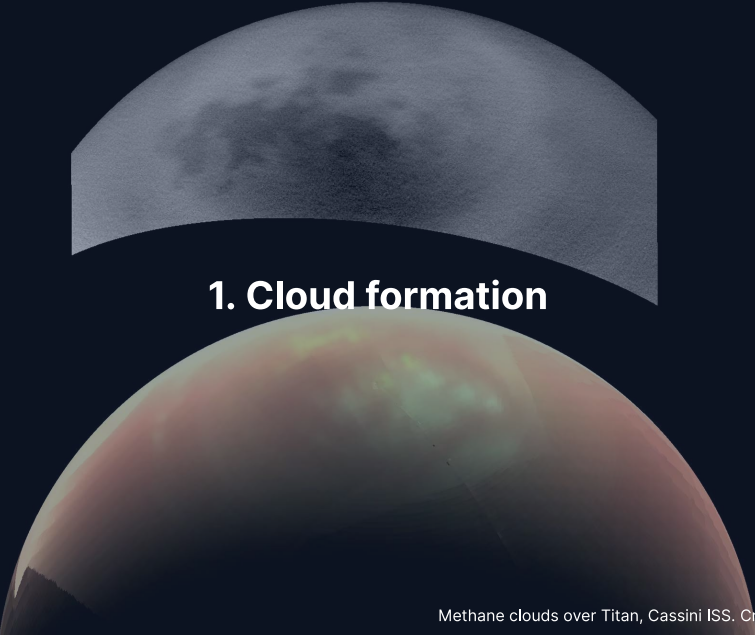
By the end of this lecture, you will be able to:

- Derive the Clausius-Clapeyron equation and predict where clouds form
- Explain circulation with the Coriolis effect, the Rossby number and geostrophic balance
- Explain the faint young Sun paradox and the carbonate-silicate thermostat

Recap: Lecture 5

- Hydrostatic equilibrium $dP = -\rho g dz$ and scale height $H = k_B T / (\mu m_u g)$ set atmospheric structure
- Radiative equilibrium and greenhouse warming set surface temperatures ($T_s > T_{eq}$)
- Convection drives the adiabatic lapse rate $\Gamma_d = g/c_p$ in the lower atmosphere

Today: What happens when vapour condenses? How do winds blow? Why has Earth's climate been stable for 4 Gyr?

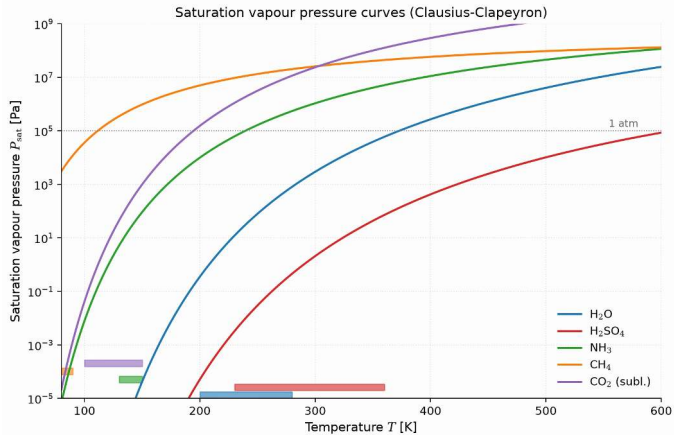


1. Cloud formation

Methane clouds over Titan, Cassini ISS. Credit: NASA/JPL-Caltech/SSI

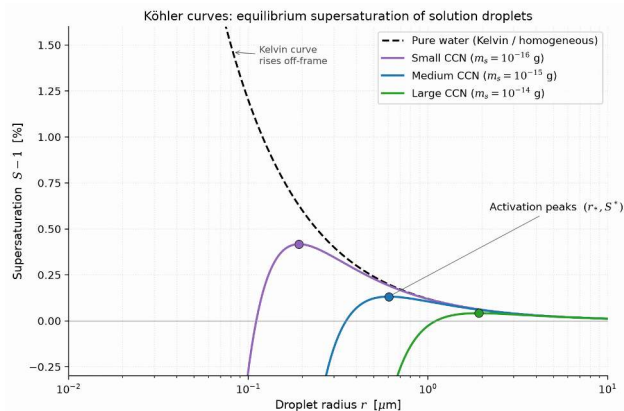
Saturation and condensation

- Every vapour has a **saturation vapour pressure** $P_{\text{sat}}(T)$ that rises roughly exponentially with temperature
- **Relative humidity** $\text{RH} = P_{\text{vapour}} / P_{\text{sat}}(T) \times 100\%$: saturated at 100%, **supersaturated** above it
- Rising air cools, P_{sat} drops, the air reaches saturation: clouds



Saturation vapour pressure against temperature for the five main condensables. Course figure.

One equation, evaluated per species and combined with the temperature profile, governs every cloud deck in the solar system.



Köhler curves: the equilibrium supersaturation of solution droplets against droplet radius.
Course figure.

Condensation needs a **seed**: homogeneous nucleation from vapour alone needs $RH \gg 100\%$; heterogeneous nucleation on dust, sea salt or soot activates at only 0.04–0.4% supersaturation.



A cumulonimbus anvil spreading at the tropopause. Credit: Eulenjäger, CC BY-SA 3.0.

Rising air cools at $\Gamma_d = g/c_p$ until it reaches its **dew point** T_d : the **lifting condensation level** (LCL) marks the **cloud base**, and above it latent heat lowers the lapse rate to about 5 K km^{-1} (9.8 dry).

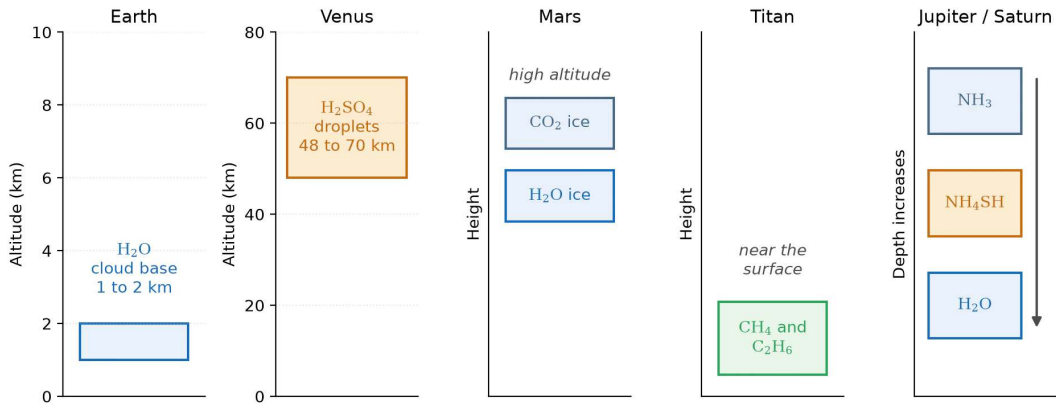
Cloud types of the solar system

What condenses depends on temperature and composition:

Body	Cloud species	Cloud location
Earth	H ₂ O	~1–2 km
Venus	H ₂ SO ₄	~48 km
Mars	CO ₂ ice, H ₂ O ice	10–100 km
Titan	CH ₄ , C ₂ H ₆	~8–30 km
Jupiter	NH ₃ , NH ₄ SH, H ₂ O	layered, 0.5–7 bar

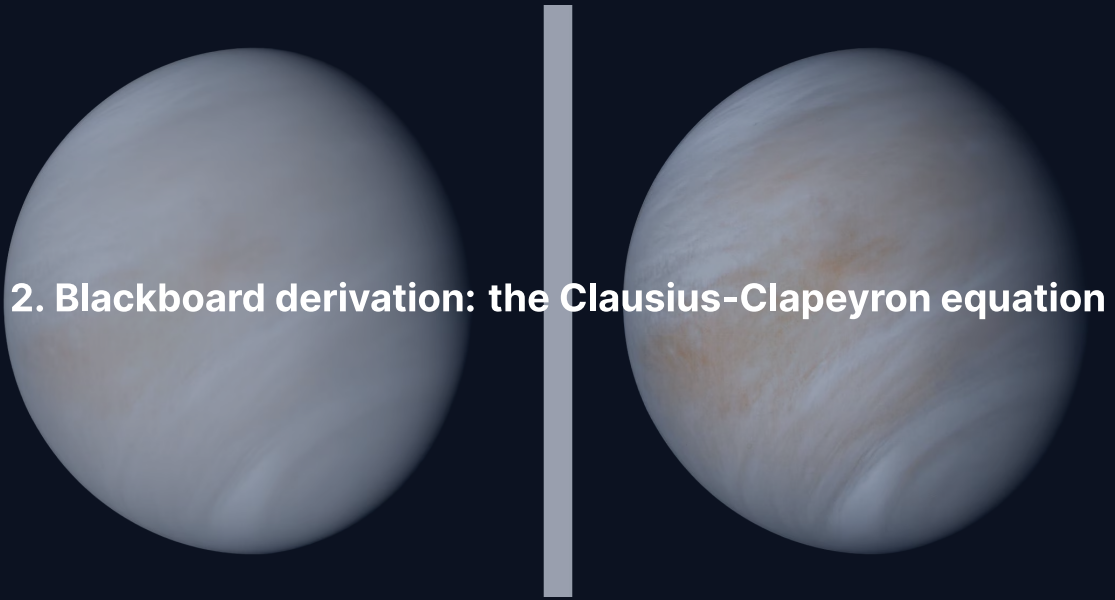
The **Clausius-Clapeyron equation** governs every cloud deck.

Which species condenses where



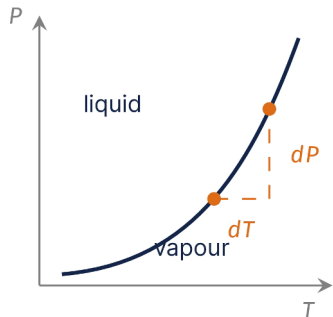
the same Clausius-Clapeyron physics in every case

Which species condenses where: H₂O on Earth, H₂SO₄ on Venus, CO₂ and H₂O ice on Mars, CH₄ and C₂H₆ on Titan, the NH₃, NH₄SH and H₂O decks on Jupiter and Saturn. Course schematic.



2. Blackboard derivation: the Clausius-Clapeyron equation

Goal and strategy



Goal: the exponential dependence of P_{sat} on temperature from phase equilibrium.

- Along the coexistence curve $g_\ell(T, P) = g_v(T, P)$, so a step (dT, dP) keeps $dg_\ell = dg_v$
- Read off the slope dP/dT and insert the latent heat
- Approximate and integrate

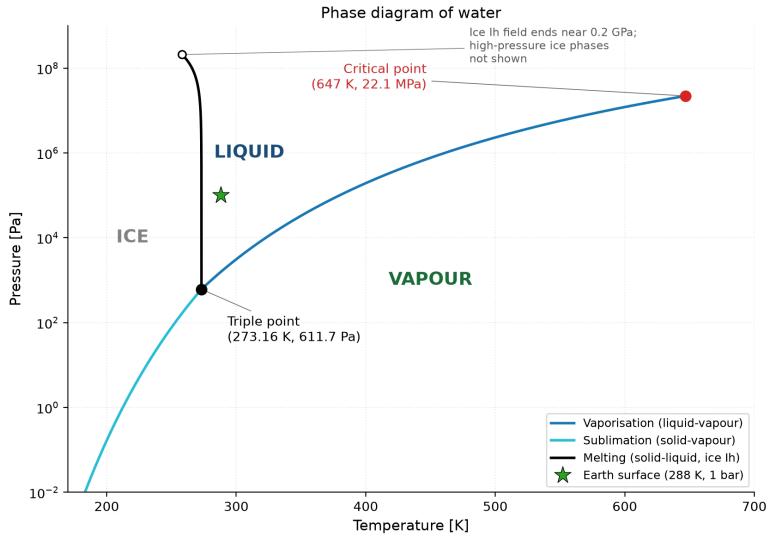
To the board.

The Clausius-Clapeyron equation

$$P_{\text{sat}}(T) = P_{\text{ref}} \exp \left[-\frac{L_v}{R_v} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right]$$

Credit: Derivation follows Catling & Kasting (2017), *Atmospheric Evolution on Inhabited and Lifeless Worlds*, Ch. 1

- P_{sat} depends **exponentially** on temperature
- The sensitivity is set by L_v/R_v (units of K): large values mean a steep curve
- The equation applies to **any** vapour to condensed-phase transition



Phase diagram of water: the liquid-vapour curve integrates Clausius-Clapeyron from the triple point to the critical point, the solid-vapour curve governs sublimation. Course figure.

Worked example: water on Earth

For water vapour:

L_v	$2.50 \times 10^6 \text{ J kg}^{-1}$
M	$0.018 \text{ kg mol}^{-1}$
$R_v = R^* / M$	$462 \text{ J kg}^{-1} \text{ K}^{-1}$
T_{ref} (triple point)	273 K
P_{ref} (triple point)	611 Pa
L_v / R_v	$\approx 5400 \text{ K}$

At $T = 293 \text{ K}$ (20°C):

$$P_{\text{sat}} = 611 \exp\left[-5400\left(\frac{1}{293} - \frac{1}{273}\right)\right] = 611 \times 3.86 \approx 2360 \text{ Pa}$$

Measured value: 2300 Pa; excellent agreement.

Condensing species in the solar system

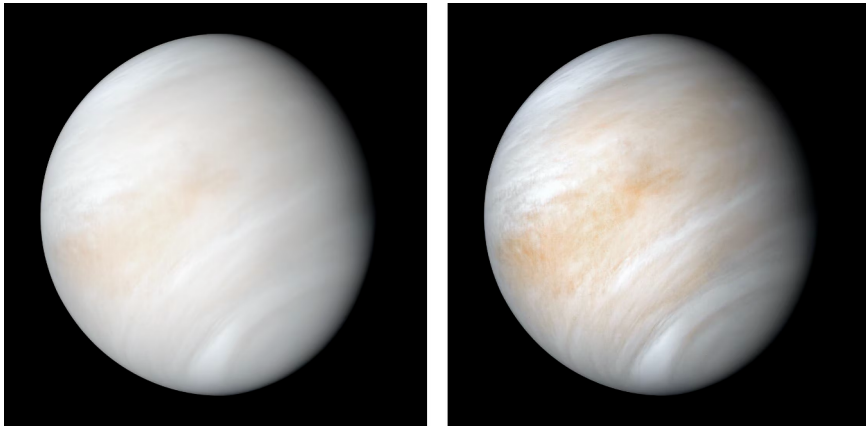
Species	L_v (kJ kg ⁻¹)	R_v (J kg ⁻¹ K ⁻¹)	L_v/R_v (K)	T_{cond} (K)	Where
H ₂ O	2500	462	5400	200–280	Earth, Mars
H ₂ SO ₄	540	85	6400	230–360	Venus
NH ₃	1370	488	2800	130–150	Jupiter, Saturn
CH ₄	510	519	980	80–90	Titan, Uranus, Neptune
CO ₂	571	189	3020	100–150	Mars

Steep L_v/R_v curves sharpen cloud bases (Venus); flat curves broaden decks (Titan).



3. Clouds of the solar system

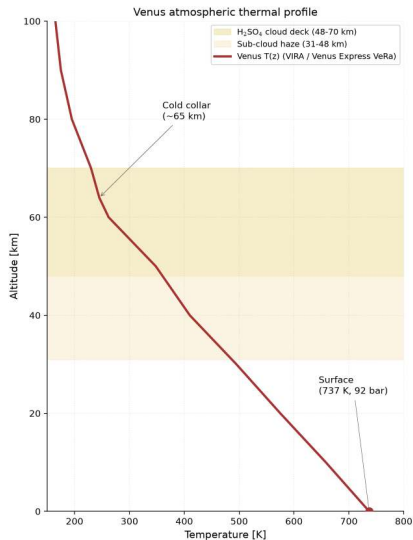
Water-ice clouds over Mars, Mars Odyssey THEMIS (2025). Credit: NASA/JPL-Caltech/ASU



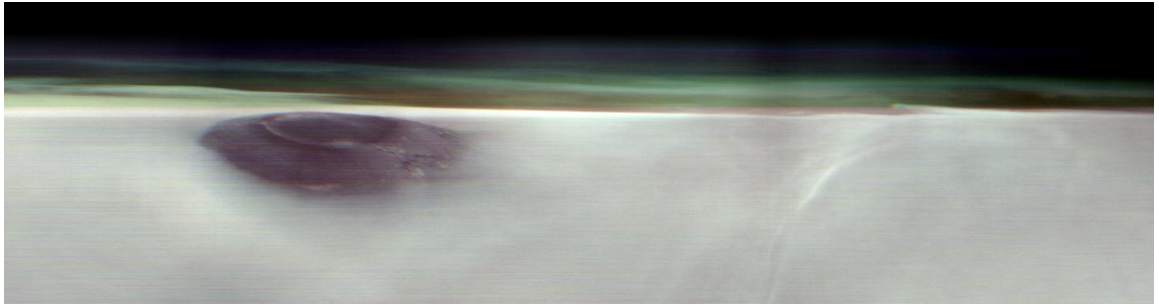
Venus in ultraviolet light, where an unidentified absorber traces the upper cloud motions.

Credit: NASA/JPL-Caltech (1974).

Concentrated H_2SO_4 clouds span 48–70 km, made from outgassed SO_2 and photolytic O ($\text{SO}_2 + \text{O} \rightarrow \text{SO}_3$, $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$); a sub-cloud haze reaches down to about 31 km.

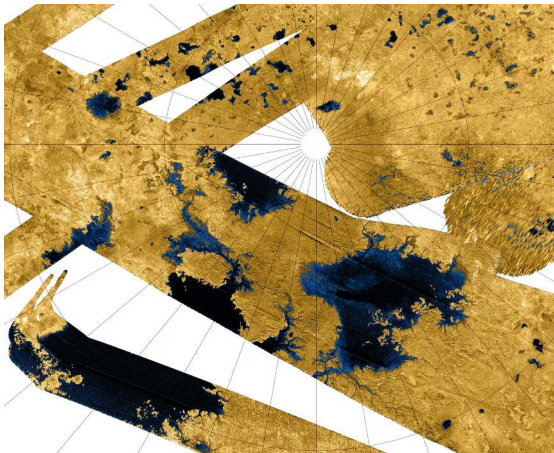


Venus thermal structure from VIRA and Venus Express VeRa: the 22 km H₂SO₄ deck sits at 48–70 km, where Clausius-Clapeyron allows it. Course figure.



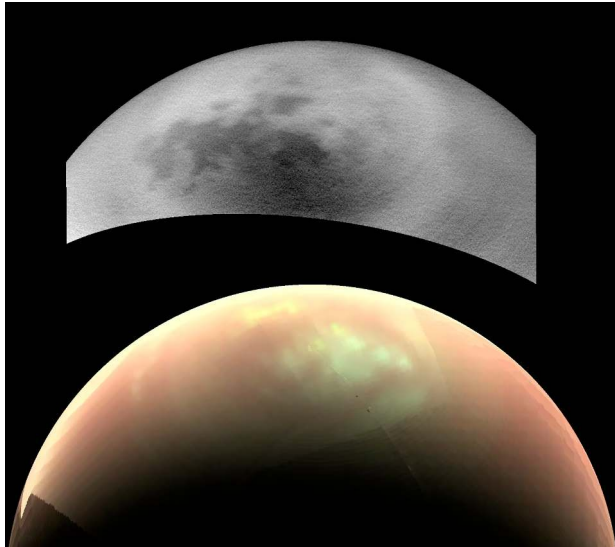
Water-ice clouds over the Tharsis limb, Mars Odyssey THEMIS, 2 May 2025. Credit: NASA/JPL-Caltech/ASU.

CO_2 ice clouds form at 50–100 km where T falls below the CO_2 frost point (148 K at the 6 mbar surface, lower aloft), H_2O ice clouds at 10–30 km over Tharsis, both seeded by mineral dust.

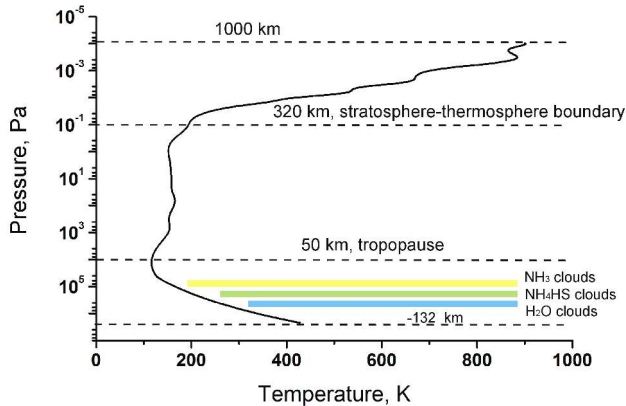


Titan's polar lakes and seas in Cassini radar. Credit: NASA/JPL-Caltech/ASI/USGS.

At 94 K and 1.5 bar methane sits near its triple point: rain carves river channels and fills polar seas such as **Kraken Mare** (~400,000 km², locally >100 m deep), paced by the 30-year seasonal cycle.



Methane clouds at Titan's mid-southern latitudes, Cassini ISS: the only active hydrological cycle beyond Earth. Credit: NASA/JPL-Caltech/SSI.



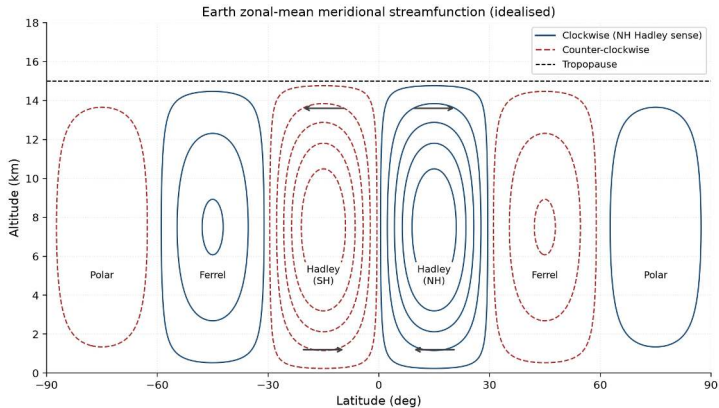
Jupiter's layered cloud structure. Credit: Wikimedia, CC BY-SA 3.0.

Clausius-Clapeyron predicts the stack: NH₃ ice at 130–150 K (0.5–1 bar), NH₄SH at 200–240 K (2–3 bar), H₂O at 270–300 K (5–7 bar), whose latent heat fuels moist convection.

A satellite image of Earth from space, showing a wide view of the planet's surface. The image captures the curvature of the Earth, with a clear blue sky and a dark, silty ocean. A prominent white cloud pattern, likely a jet stream, stretches across the upper half of the frame. Below it, the landmasses of North and South America are visible, with various cloud formations over the continents and surrounding waters. The overall scene is a high-contrast, high-resolution view of the Earth's atmosphere and surface features.

4. Atmospheric dynamics

Jet-stream cirrus from orbit. Credit: NASA/JSC



Idealised zonal-mean overturning streamfunction of Earth's troposphere, after Held & Hou (1980). Course figure.

Equatorial heating lifts air that flows poleward aloft, sinks near 30° latitude over the subtropical deserts and returns as the trade winds: the tropical heat transport loop.

The Coriolis effect

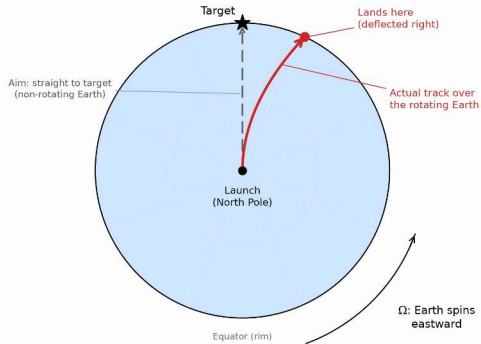
Moving air is deflected right in the Northern Hemisphere, left in the Southern.
The **Coriolis parameter**:

$$f = 2\Omega \sin \phi$$

- Earth: $\Omega = 7.29 \times 10^{-5} \text{ rad s}^{-1}$
- Mid-latitudes ($\phi = 45^\circ$): $f \approx 1.03 \times 10^{-4} \text{ s}^{-1}$
- Equator: $f = 0$, no deflection

Winds circulate around pressure centres instead of blowing from high to low.

Geometric origin of the Coriolis deflection
(view from above the North Pole)



The Earth turns east under the flight, so the track bends to the right of the direction of travel (Northern Hemisphere; it bends left in the Southern).

Coriolis deflection of a polar launch seen from above the North Pole: an apparent force from the planet rotating beneath a straight path. Course figure.

The Rossby number

Does planetary rotation dominate a given flow?

$$Ro = \frac{U}{fL}$$

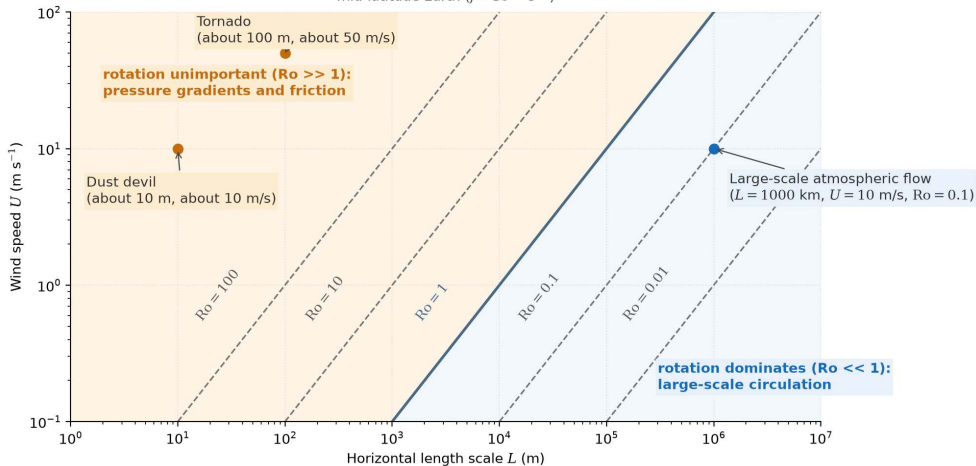
$Ro \ll 1$: rotation dominates: large-scale circulation, jet streams (Earth:
 $U \sim 10 \text{ m s}^{-1}$, $L \sim 1000 \text{ km}$, $Ro \sim 0.1$)

$Ro \gg 1$: rotation unimportant: convective plumes, tornadoes

The Rossby number sets the dynamic regime and the number of circulation cells.

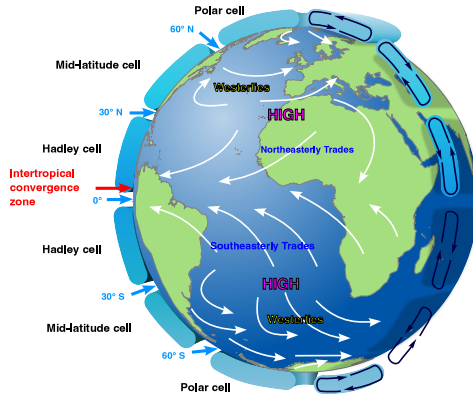
The Rossby number sorts flows by the role of rotation

mid-latitude Earth ($f = 10^{-4} \text{ s}^{-1}$)



Flow regimes sorted by $Ro = U/(fL)$ for mid-latitude Earth: large-scale circulation where rotation dominates, dust devils and tornadoes where pressure gradients and friction govern.

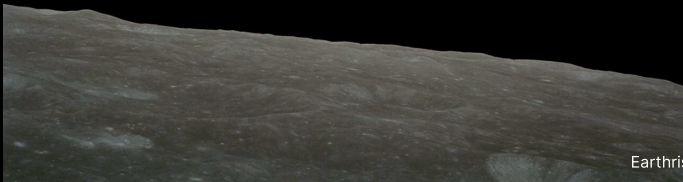
Course figure.



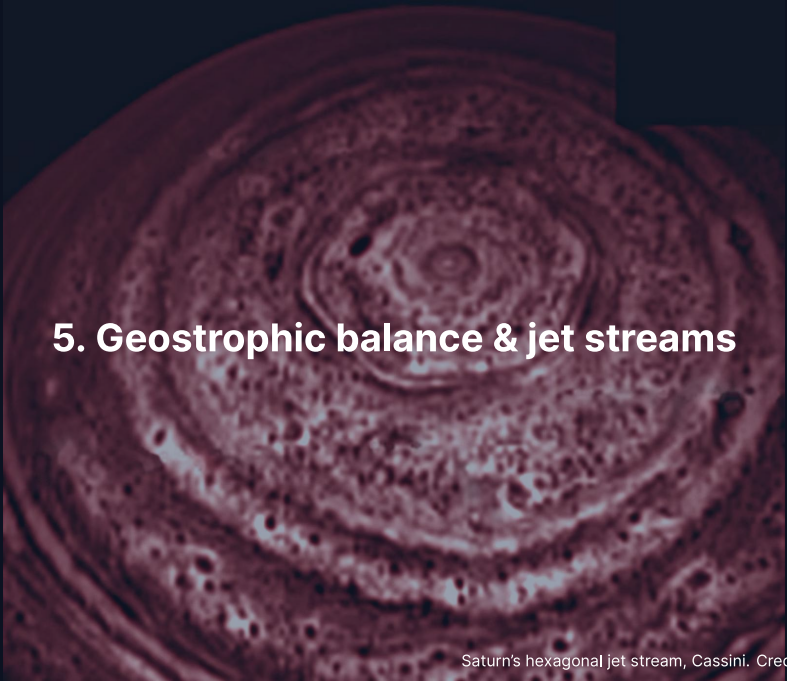
Earth's three-cell circulation, the pattern of a moderate rotator: Hadley, mid-latitude (Ferrel) and polar cells. Credit: Wikimedia Commons, public domain.

Slow rotators (Venus, Titan): one equator-to-pole cell per hemisphere; Earth: three cells (Hadley, Ferrel, polar); **fast rotators** (Jupiter, Saturn): many cells that appear as banded jets.

Break



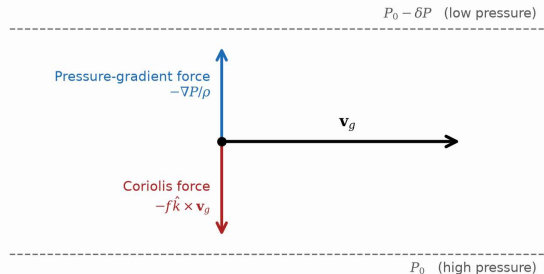
Earthrise, Apollo 8 (1968). Credit: NASA

A detailed view of Saturn's atmosphere showing a prominent hexagonal jet stream. The hexagon is a large, six-sided cloud pattern with a smaller hexagon inside it. The surrounding atmosphere shows various cloud bands and features. The image is in a reddish-brown color scheme.

5. Geostrophic balance & jet streams

Saturn's hexagonal jet stream, Cassini. Credit: NASA/JPL-Caltech/SSI

Geostrophic balance (Northern Hemisphere)



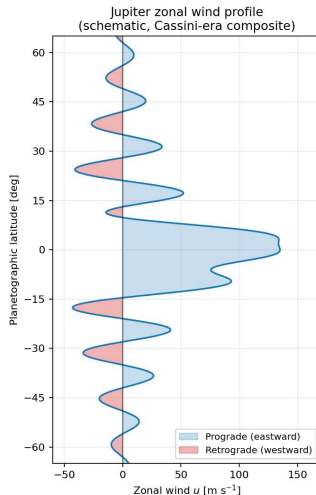
Geostrophic balance between the pressure-gradient force and the Coriolis force. Course figure.

At $R_o \ll 1$, $f\hat{k} \times \mathbf{v}_g = -\nabla P/\rho$: geostrophic winds blow **parallel to isobars**, cyclonically around lows and anticyclonically around highs.

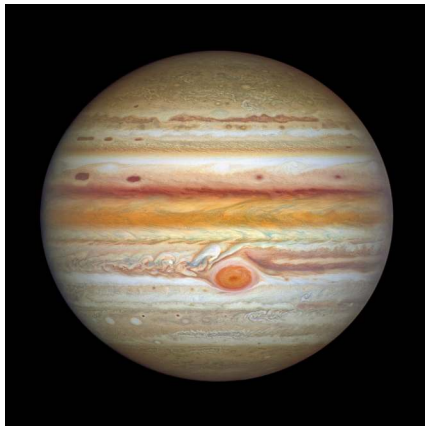


Jet-stream cirrus from orbit. Credit: NASA JSC, public domain.

The thermal wind, $\partial \mathbf{v}_g / \partial z \propto \hat{k} \times \nabla T$: the equator-to-pole temperature gradient drives $30\text{--}70 \text{ m s}^{-1}$ jets aloft, subtropical near 30° and polar near 60° , that steer mid-latitude weather.



Jupiter's cloud-top zonal winds: alternating prograde and retrograde jets peak near 180 m s^{-1} and reach thousands of kilometres deep. Course figure, after García Melendo & Sánchez-Lavega (2001).

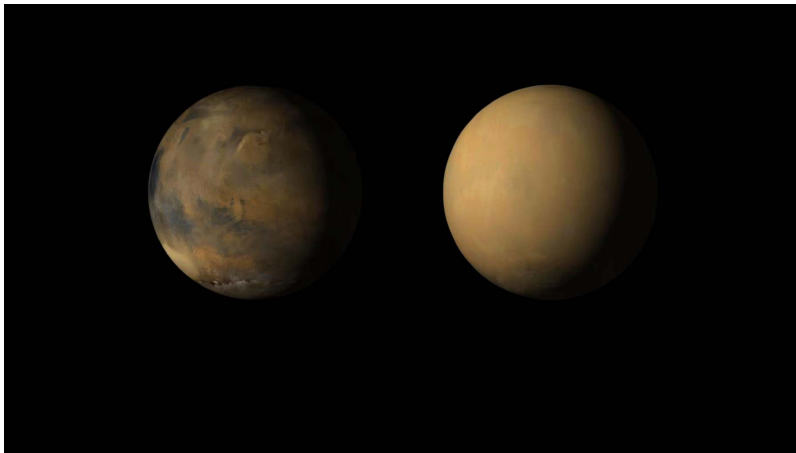


Jupiter's global map from the Hubble OPAL programme. Credit: NASA/ESA/STScI.

Bright **zones** are anticyclonic upwelling with cool ammonia ice tops, dark **belts** cyclonic downwelling exposing warmer layers; zonal jets flank the boundaries (Jupiter ~ 180 , Saturn $\sim 400 \text{ m s}^{-1}$); Juno gravity puts Jupiter's jet decay depth near 1800 km.

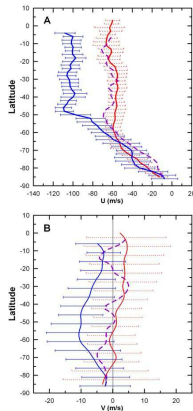
A close-up, high-resolution image of Jupiter's Great Red Spot, a massive, oval-shaped storm system. The storm is characterized by a reddish-orange center surrounded by swirling white and light brown clouds. The image is presented as a semi-circular arc against a dark blue background.

6. Weather & storms in the solar system



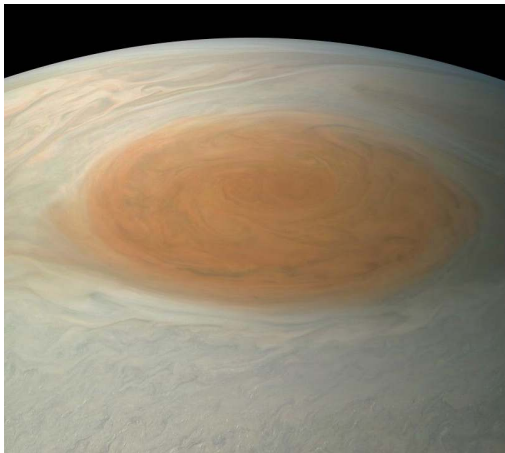
The 2018 global dust storm on Mars. Credit: NASA/JPL-Caltech/MSSS.

Dust lofted to 10–30 km absorbs sunlight, heats the air and strengthens the winds that loft more dust, a **positive feedback**; seasonal CO₂ frost on the caps (25–30% of the atmosphere) swings the surface pressure between about 5 and 7 mbar.



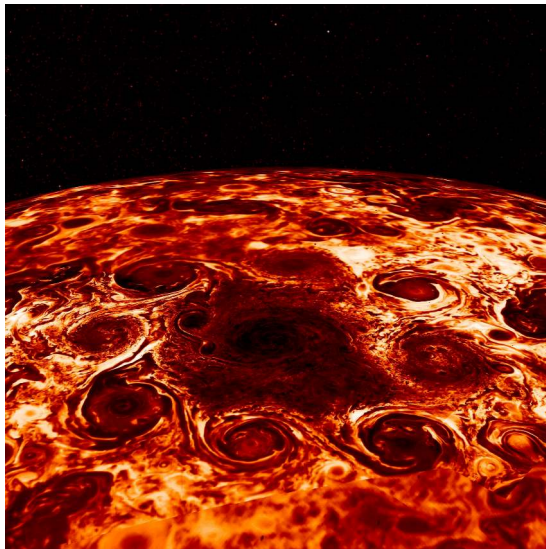
Venus cloud-level winds from VIRTIS on Venus Express: zonal (A) and meridional (B) speed against latitude. Sánchez-Lavega et al. (2008).

Cloud-top winds of $\sim 100 \text{ m s}^{-1}$ circle Venus in about 4 days, 60 times faster than the surface: **thermal tides** pump momentum upward and **planetary waves** carry it equatorward (Lecture 9).

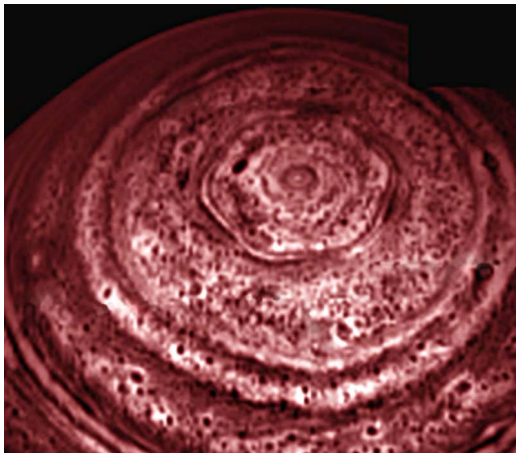


The Great Red Spot from Juno. Credit: NASA/JPL-Caltech/SwRI/MSSS.

An anticyclone larger than Earth, recorded since about 1830, with peripheral winds near 120 m s^{-1} between opposing jets, fed by merging vortices and deep H_2O latent heat, and slowly shrinking in longitude.



Jupiter's north polar cyclone cluster from Juno JIRAM: a central cyclone ringed by eight circumpolar cyclones in a stable octagon. Credit: NASA/JPL-Caltech/SwRI/ASI/INAF/JIRAM.



Saturn's north polar hexagon, Cassini 2013. Credit: NASA/JPL-Caltech/SSI.

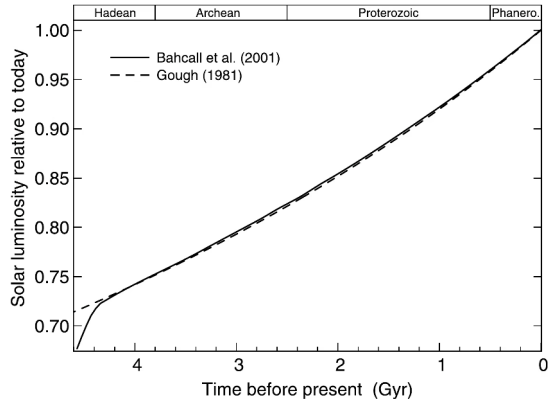
A stationary Rossby wave with azimuthal wavenumber $m = 6$ locked to the jet at $\sim 78^\circ\text{N}$, about 30,000 km across, seen by Voyager and Cassini; Saturn also erupts Great White Storms every ~ 30 years (last in 2010).



Neptune's Great Dark Spot, Voyager 2 (1989): Neptune radiates 2.6 times the sunlight it absorbs, and that internal heat drives $\sim 400 \text{ m s}^{-1}$ winds. Credit: NASA/JPL-Caltech.

An artist's impression of a Snowball Earth, showing a frozen planet with a thin blue ring of liquid water at the equator and a dark blue polar region. The text "7. Climate evolution & the faint young Sun" is overlaid in white.

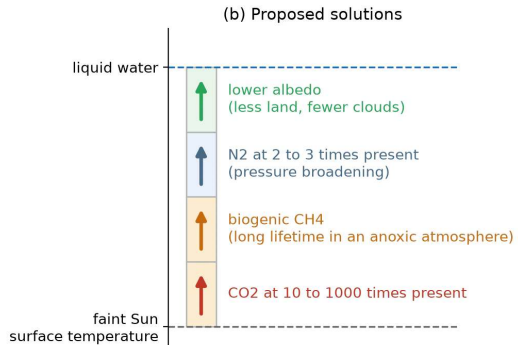
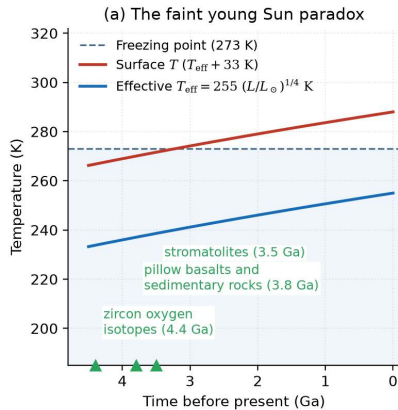
7. Climate evolution & the faint young Sun



Solar luminosity through geologic time: the Sun was about 30% fainter ($0.71 L_{\odot}$) at $t = 0$.

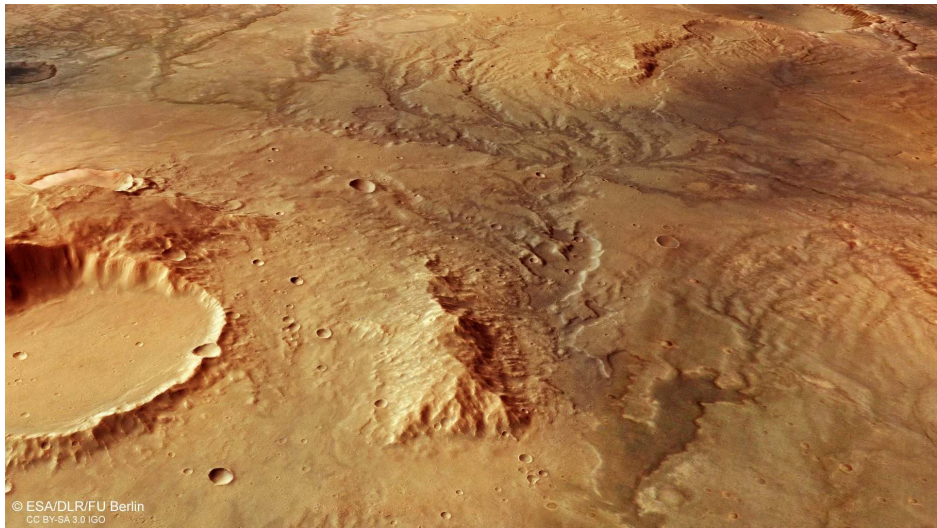
Reproduced from Feulner (2012).

$L(t)/L_{\odot} \approx [1 + \frac{2}{5}(1 - t/t_{\odot})]^{-1}$ gives $T_{\text{eff}} \approx 234$ K for the early Earth, yet zircons (4.4 Ga), pillow basalts (3.8 Ga) and stromatolites (3.5 Ga) record liquid water: the **faint young Sun paradox**.

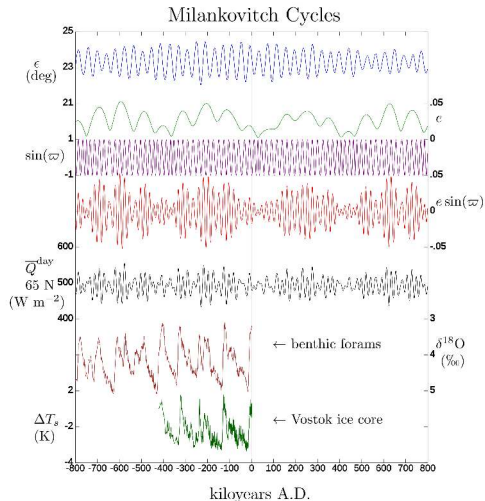


(a) Equilibrium and greenhouse temperatures under the fainter Sun against the geological record of liquid water; (b) the proposed warming contributions. Course figure.

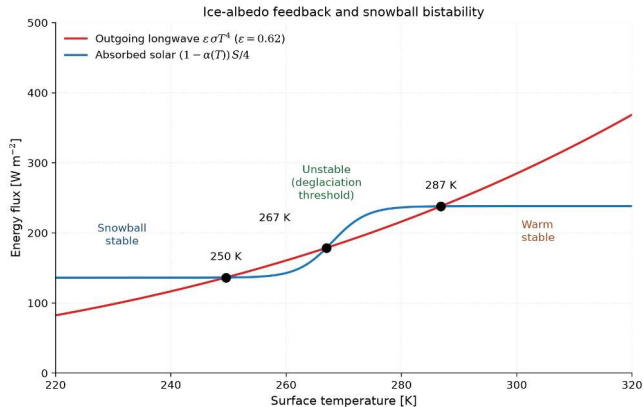
Solutions: CO_2 at 10 to 1000 times present, biogenic CH_4 , N_2 at 2 to 3 times present, and a lower planetary albedo (less land, fewer clouds).



Noachian valley networks, Mars Express HRSC: warm, wet early Mars (>3.7 Ga) is an open paradox, since pure CO₂ condenses into bright clouds. Credit: ESA/DLR/FU Berlin, CC BY-SA 3.0 IGO.



Milankovitch cycles over 800 kyr: obliquity, eccentricity and precession pace the glacial cycles, though the 100 kyr rhythm needs non-linear feedbacks. Credit: Wikimedia Commons, CC BY-SA.



Energy-balance bistability of the ice-albedo feedback: stable states at 250 K and 287 K, an unstable threshold at 267 K. Course figure.

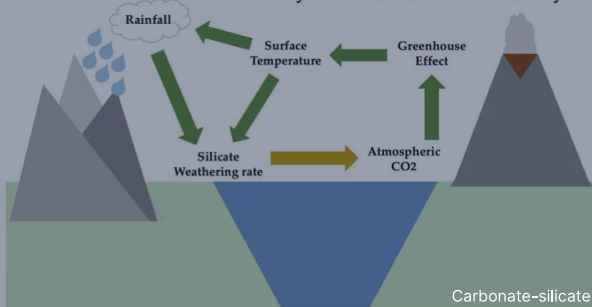
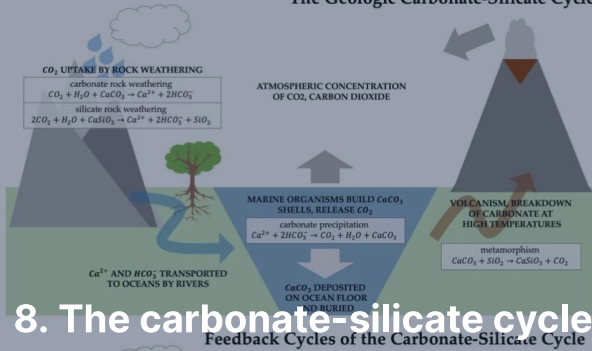
The **ice-albedo feedback** is positive: cooling expands reflective ice, and past ~30° latitude it runs away into the snowball state; two stable climates sit either side of the 267 K threshold.



Snowball Earth, artist's impression. Credit: Oleg Kuznetsov, CC BY-SA 4.0.

Two Neoproterozoic glaciations (~717 and 635 Ma) left tropical glacial deposits at surfaces below -50°C ; ice shut down silicate weathering until volcanic CO_2 built up over 10–100 Myr and melted it (Hoffman et al. 2017).

The Geologic Carbonate-Silicate Cycle



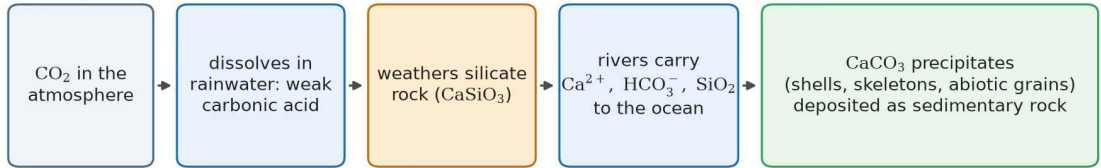
The Urey reaction

Silicate weathering draws atmospheric CO₂ into mineral carbonates:



- CO₂ dissolves in rain as carbonic acid (H₂CO₃) and weathers surface silicates
- Rivers carry Ca²⁺ and HCO₃⁻ to the oceans, where carbonate (CaCO₃) precipitates
- Carbon moves from atmosphere and ocean into the crust, until subduction and volcanism return it

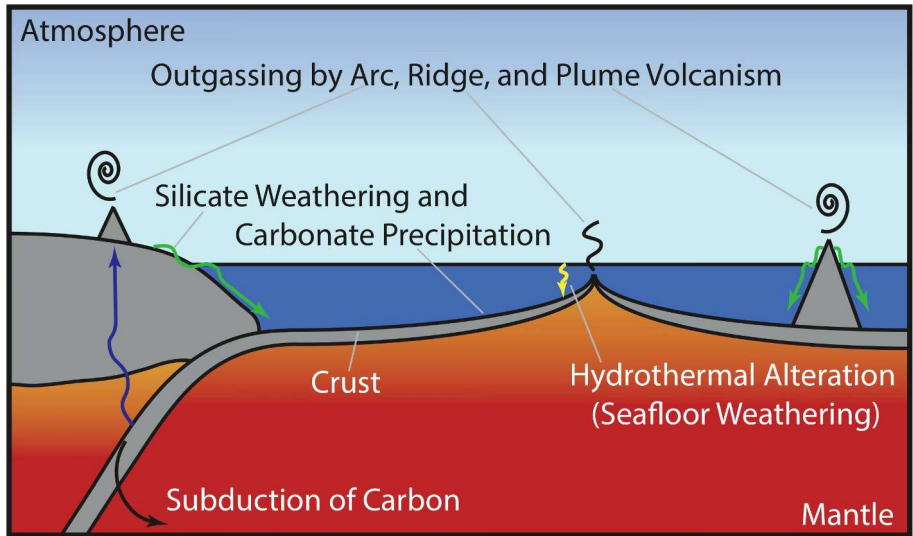
The Urey reaction pathway



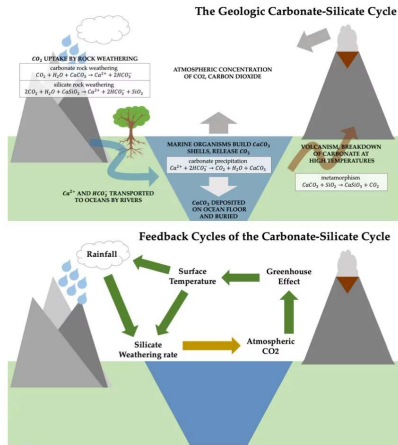
net effect: CO₂ drawn out of the atmosphere into carbonate rock, the long-term carbon sink



The Urey reaction pathway from rain to carbonate rock: CO₂ drawn out of the atmosphere into the long-term carbon sink. Course schematic.



The carbonate-silicate cycle on the modern Earth: weathering into oceanic carbonates, subduction, and metamorphic CO_2 returned by arc and plume volcanism. Credit: Foley (2024).



The carbonate-silicate cycle. Credit: Wikimedia, CC BY-SA 4.0.

Subducted carbonate decomposes at high T and P and **volcanic outgassing** returns the CO_2 : the long-term steady state balances the weathering sink against the volcanic source.

The negative feedback thermostat

Silicate weathering rates rise exponentially with temperature:

$$W(T) = W_0 \exp\left(\frac{T - T_0}{T_e}\right)$$

- **Warming:** more rain and faster reactions draw CO_2 down and cool the planet
- **Cooling:** weathering slows and volcanic CO_2 accumulates
- With $T_e \approx 10$ K, doubling the weathering rate needs ~ 7 K of warming

Perturbations damp on ~ 0.5 Myr timescales, maintaining liquid water for >4 Gyr.

Source: Walker et al. (1981); Pierrehumbert (2010)

The thermostat needs both ingredients

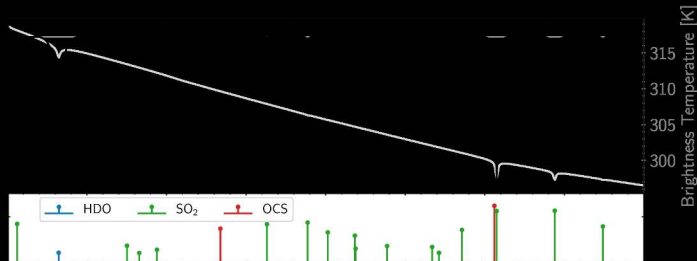
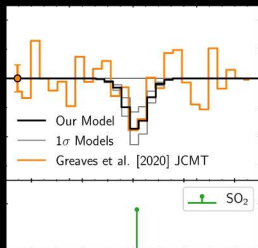
	Liquid water: yes	Liquid water: no
Active volcanism: yes	Earth Thermostat works CO ₂ regulated over Gyr	Venus Weathering stops, volcanic CO ₂ accumulates: 92 bar
Active volcanism: no	Mars (after about 3 Ga) Source broken, weathering and escape draw CO ₂ down: 6 mbar	No cycle No water or volcanism Thermostat fails

The thermostat needs liquid water and volcanism: Earth has both, Venus lost its water, Mars its volcanism. Course schematic.

Venus lost the sink (no water, volcanic CO₂ reached 92 bar); Mars lost the source (volcanism ceased ~3 Ga, weathering and escape left ~6 mbar): habitability needs active geological cycling, not sunlight alone.

The image features two spheres side-by-side against a dark blue background. The sphere on the left is a realistic representation of Mars, showing its reddish-brown surface with darker patches and polar ice caps. The sphere on the right is a uniform, hazy, light-tan color, representing the planet completely obscured by a global dust storm. Centered over the space between these two spheres is the section title.

9. Recent advances & outlook



The reported 267 GHz Venus phosphine feature is fit by mesospheric SO₂, a controversy that NASA's DAVINCI probe will test in situ. Lincowski et al. (2021).

Summary

1. The Clausius-Clapeyron equation predicts saturation vapour pressures and cloud bases on every planet
2. Rotation sets the circulation: Coriolis, Rossby number, geostrophic winds and jets, from single Hadley loops to banded jets
3. The faint young Sun paradox is resolved by greenhouse warming and lower albedo; carbonate-silicate weathering is a ~ 0.5 Myr climate thermostat

Next: Lecture 7

Questions?

Next: Lecture 7, Planetary surfaces: geology, geomorphology & geophysics

Mon 28 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

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